Student Academic Performance Prediction

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1. **Problem Description**...3

2. **Architecture** 4

3. **Models model** 5

4. **Evaluation metrics used**...6

5. **Preliminary Results and Visualizations**...7

6. **References** 8

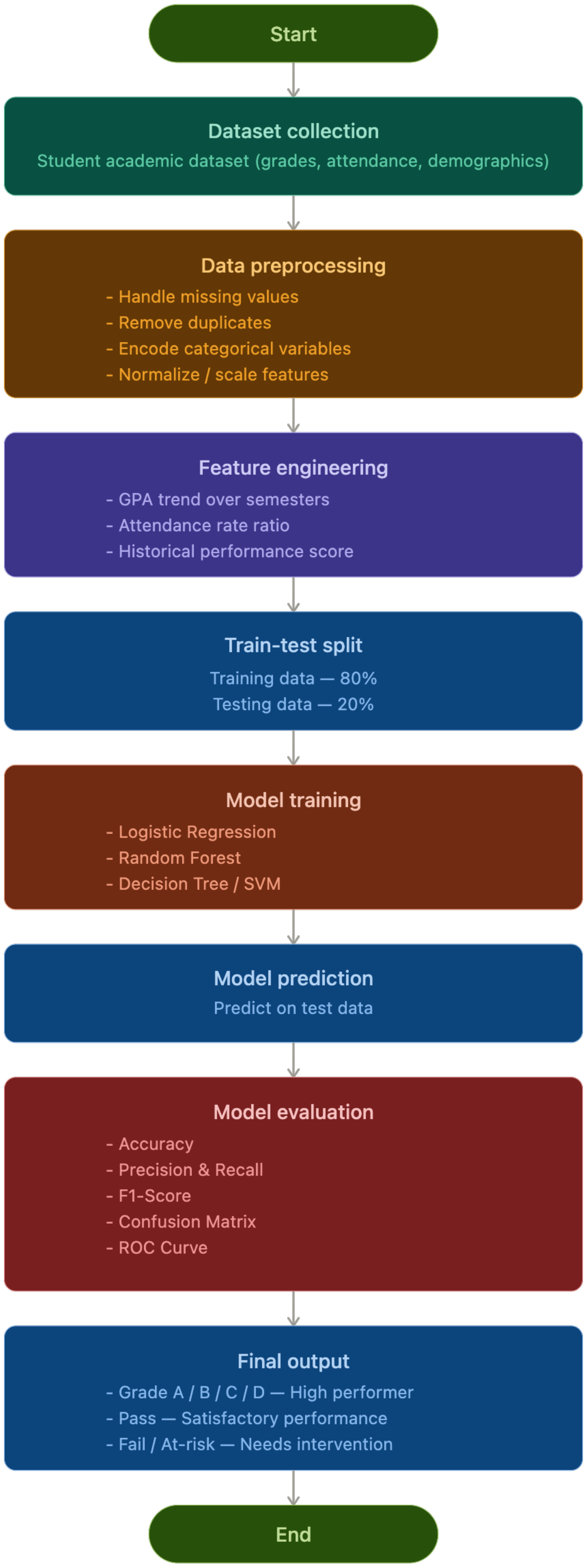
1. Problem Description

Educational institutions often struggle to identify students who are at risk of poor academic performance in a timely manner. Traditional methods rely on manual analysis of student records, which is time-consuming, error-prone, and inefficient for large datasets.

The Student Academic Performance Prediction System aims to solve this problem by using machine learning techniques to automatically analyze student data and predict academic outcomes. The system processes various factors such as attendance, marks, study habits, and other relevant features to build predictive models.

By applying data preprocessing, feature selection, and classification/regression algorithms, the system provides accurate predictions and insights. This helps educators take early interventions, improve student performance, and make data-driven decisions.

2. Architecture



3. Models Used(Applied/ Planned)

1. **Linear Regression**
   A supervised learning algorithm used to predict continuous values such as student marks. It finds a relationship between input features (like attendance, study time) and the output (performance score).

2. **Logistic Regression**
   Used for classification tasks, such as predicting whether a student will *pass or fail*. It outputs probabilities and classifies results into categories.

3. **K-Nearest Neighbors (KNN)**
   A simple algorithm that classifies a student based on the performance of similar students (nearest neighbors). It works well for pattern recognition in smaller datasets.

4. **Decision Tree**
   A tree-structured model that makes decisions based on conditions (e.g., attendance > 75%). It is easy to understand and interpret.

5. **Random Forest (Planned)**
   An advanced model that uses multiple decision trees to improve accuracy and reduce overfitting. It provides more reliable predictions.

6. **Support Vector Machine (SVM) (Planned)**
   A powerful classification algorithm that finds the best boundary to separate different classes (e.g., pass/fail). It performs well on complex datasets.

4. Evaluation metrics used

Accuracy – Measures overall correct predictions out of total predictions.

Precision – Shows how many predicted positive results are actually correct.

Recall – Measures how well the model identifies all actual positive cases.

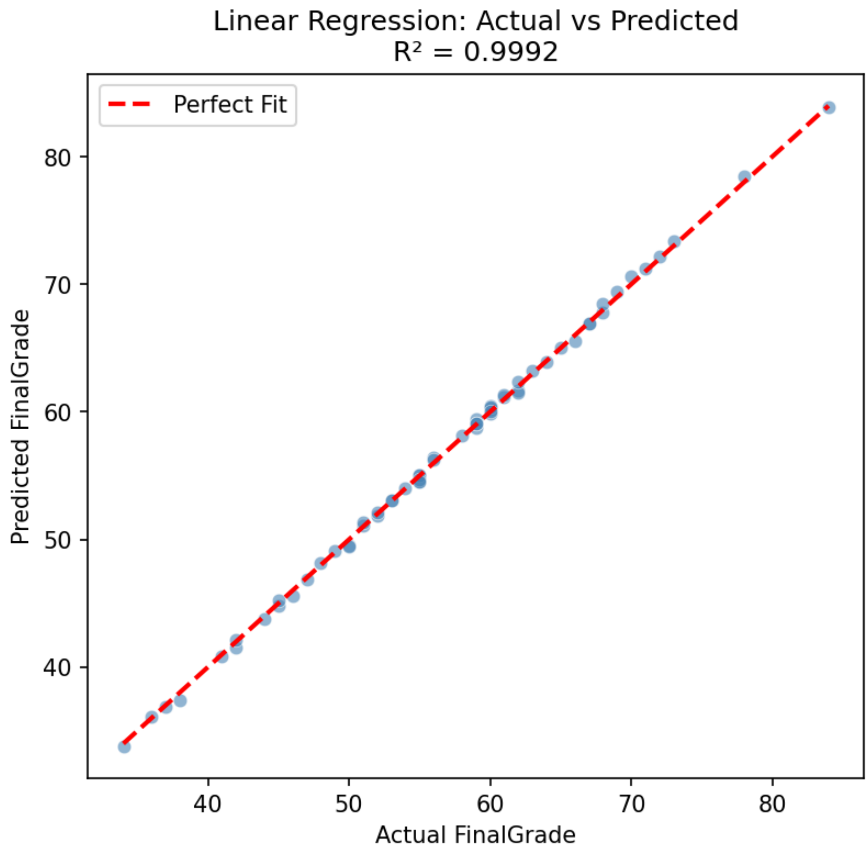
F1-Score – Balance between precision and recall.

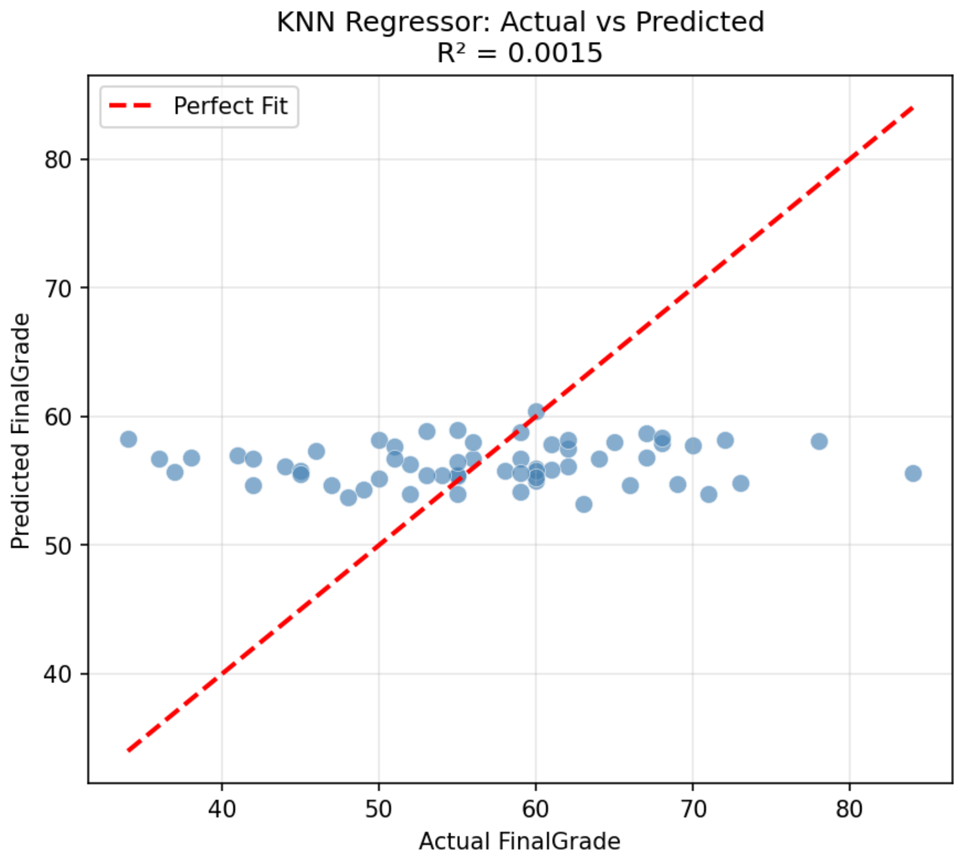
MAE (Mean Absolute Error) – Average difference between actual and predicted values.

MSE (Mean Squared Error) – Similar to MAE but gives more weight to larger errors.

Confusion Matrix – Table used to evaluate model performance showing TP, FP, TN, FN.

5. Preliminary Results and Visualizations





6. References:

Books:
Python Machine Learning – Sebastian Raschka
Introduction to Machine Learning – Ethem Alpaydin

Online Resources:
W3Schools – https://www.w3schools.com/
Google Scholar – https://scholar.google.com/